

MOVIE SUCCESS PREDICTOR

Exploratory Data Analysis (EDA) Report

IMDb 5000 Movie Dataset

1. Introduction

The Movie Success Predictor project uses the IMDb 5000 Movie Dataset to explore factors associated with movie performance. The EDA notebook analyzes 5,043 movie records across 28 attributes. The analysis focuses on movie characteristics such as budget, gross revenue, IMDb score, audience engagement, reviews, genre, content rating, release year, and director information.

The purpose of Exploratory Data Analysis is to understand the structure and quality of the dataset before applying statistical analysis or predictive models. The notebook uses descriptive statistics, distributions, trend analysis, bar charts, scatter plots, and a correlation heatmap to identify patterns and relationships in the movie data.

2. Dataset Overview

Item	Finding
Number of movies	5,043
Number of attributes	28
Numerical attributes	16 (float64/int64 in the notebook output)
Categorical/text attributes	12 object columns
Main performance variables	gross revenue and IMDb score
Main analysis relationship	Budget vs Gross Revenue

The dataset contains both numerical and categorical information. Numerical variables include budget, gross revenue, duration, IMDb score, review counts, vote counts, Facebook likes, and release year. Categorical variables include color, director, actors, genres, language, country, content rating, and movie title.

3. EDA Workflow

- Import Python libraries including Pandas, NumPy, Matplotlib, and Seaborn.
- Load the IMDb 5000 dataset and inspect its shape and first records.
- Check data types and dataset information.
- Calculate descriptive statistics for numerical variables.
- Identify missing values and their distribution across columns.
- Study numerical distributions using histograms with KDE.
- Analyze release trends, genres, ratings, directors, and top movies.
- Use scatter plots to examine relationships between important variables.
- Use a correlation heatmap to summarize relationships among numerical features.

The notebook is therefore structured as a complete EDA stage: it describes the data, checks data quality, explores individual variables, and investigates relationships between variables.

4. DATA QUALITY AND DESCRIPTIVE ANALYSIS

4.1 Missing Value Analysis

The EDA notebook reports missing values in several columns. The missing-value check shows that the dataset used by this notebook is not completely cleaned at the time of EDA. The largest gaps occur in gross revenue, budget, aspect ratio, content rating, and plot keywords.

Column	Missing values
gross	884
budget	492
aspect_ratio	329
content_rating	303
plot_keywords	153
title_year	108
director_name	104
director_facebook_likes	104
num_critic_for_reviews	50
duration	15

Other columns also contain smaller numbers of missing values, including color, actor information, language, country, and review-related variables. Missing values are important because they can affect summary statistics, visualizations, correlation calculations, and predictive modeling. The project should therefore handle them during preprocessing before final model training.

4.2 Descriptive Statistics

The descriptive statistics show substantial variation across the movie variables. The average movie duration is approximately 107.2 minutes, with a median of 103 minutes. IMDb scores have a mean of 6.44, a median of 6.6, and a maximum of 9.5. Gross revenue has a mean of about \$48.47 million among the available records, while the median is about \$25.52 million.

Variable	Mean	Median	Maximum
Duration (minutes)	107.20	103	511
IMDb score	6.44	6.60	9.50
Gross revenue	\$48.47M	\$25.52M	\$760.51M
Budget	\$39.75M	\$20.00M	\$12.22B
Number of voted users	83,668	34,359	1,689,764

The large differences between means and medians for variables such as gross, budget, and audience engagement indicate that these variables have highly uneven distributions. A small number of movies have very large values, so these variables should be interpreted carefully and may require transformation or robust statistical methods in later modeling.

5. VISUAL EXPLORATION AND RELATIONSHIPS

5.1 Numerical Distributions

The notebook creates histograms with KDE curves for duration, budget, gross revenue, IMDb score, number of voted users, review counts, and Facebook likes. These plots are used to understand the spread, concentration, and skewness of the numerical variables. In particular, budget, gross revenue, vote counts, and Facebook-like variables contain very large maximum values compared with their medians, showing that the distributions are strongly influenced by a smaller group of high-value movies.

5.2 Movie Release Trend

A line chart is used to show the number of movies released by year. This provides a historical view of the dataset and helps identify periods in which more movies are represented. The chart is descriptive; it should not be interpreted as the total number of movies produced by the film industry because the dataset is a sample rather than a complete industry census.

5.3 Genre Analysis

The notebook identifies the top 10 genre combinations by frequency and displays them in a horizontal bar chart. It also compares average gross revenue across those top genres. This allows the project to examine both the distribution of movie types and whether average commercial performance differs across frequently occurring genre groups.

5.4 Budget and Gross Revenue

A scatter plot compares production budget with gross revenue. This is one of the central analyses in the project because the project investigates whether movies with larger production budgets tend to generate larger box-office collections. The scatter plot is useful for identifying the overall direction of the relationship, the spread of observations, and unusually high or low revenue values. A relationship in the plot should be interpreted as an association rather than proof that budget alone causes revenue.

5.5 IMDb Score and Audience Engagement

The notebook compares IMDb score with number of voted users. It also examines budget versus IMDb score. These plots help distinguish commercial variables from audience-rating variables and show whether financial investment is closely associated with rating outcomes.

5.6 Content Rating and Directors

Bar charts are used to compare average IMDb score by content rating and to identify directors with the largest number of movies in the dataset. These analyses add categorical context to the numerical analysis and demonstrate that movie performance can be studied from several dimensions rather than from budget and revenue alone.

6. CORRELATION ANALYSIS AND KEY EDA FINDINGS

6.1 Correlation with IMDb Score

The notebook calculates correlations among duration, budget, gross, IMDb score, number of voted users, user reviews, critic reviews, and movie Facebook likes. The strongest positive correlation with IMDb score among the selected variables is with number of voted users (0.411). Other positive correlations are observed with critic reviews (0.305), user reviews (0.292), duration (0.262), movie Facebook likes (0.247), and gross revenue (0.198).

Variable	Correlation with IMDb score
Number of voted users	0.411
Number of critic reviews	0.305
Number of user reviews	0.292
Duration	0.262
Movie Facebook likes	0.247
Gross revenue	0.198
Budget	0.031

These values indicate that budget has a very small linear correlation with IMDb score in this analysis, while audience voting and review-related variables show stronger positive associations. Correlation does not establish causation and does not by itself determine which variables should be used as the final prediction target or model features.

6.2 Top Movies Identified by the Notebook

The EDA also extracts the top 10 movies by gross revenue and the top 10 by IMDb score. The highest gross-revenue record shown is Avatar at approximately \$760.5 million. The highest IMDb score shown is 9.5 for Towering Inferno in this dataset output. The top-movie tables are descriptive outputs and are useful for demonstrating how the dataset can be filtered and ranked.

6.3 Overall Findings

- The dataset contains 5,043 movies and 28 attributes, providing a broad mixture of movie, financial, audience, and categorical information.
- Missing values are present in multiple attributes, with gross revenue and budget among the variables having substantial missing counts.
- Budget and gross revenue show large ranges and extreme maximum values, so distribution shape and outliers should be considered during modeling.
- IMDb score has a mean of 6.44 and a median of 6.6 in the analyzed data.
- The correlation analysis shows a stronger positive association between IMDb score and number of voted users than between IMDb score and budget.
- The EDA uses multiple visualization types to examine distributions, trends, category frequencies, financial relationships, audience engagement, and correlations.

7. Conclusion

The EDA provides a structured understanding of the IMDb 5000 Movie Dataset before predictive modeling. It identifies data-quality issues, summarizes important numerical variables, and explores relationships between budget, gross revenue, IMDb score, audience engagement, and categorical movie characteristics. The findings show why preprocessing is necessary and why movie success should not be represented by a single factor alone. The next stage of the project can use the cleaned data and a clearly defined prediction target to build and evaluate suitable statistical or machine-learning models.

Source basis: EDA.ipynb supplied for this report. The report preserves the variables, outputs, and analyses present in the notebook.